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A self-supervised contrastive learning approach for latent fingerprint identification

Andre Nobrega [a,b,*], Ilan Theodoro [a], Pascual Figueroa [b], Alexandre Xavier Falcão [a]

[a] Institute of Computing - University of Campinas (Unicamp), Av. Albert Einstein, 1251, Campinas, 13083-852, Brazil
[b] Griaule Biometrics, Av. Dr. Romeu Tortima, 1448, Campinas, 13083-897, Brazil

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ABSTRACT

Latent fingerprints are challenging to identify due to low quality, partial impressions, and noise. This paper proposes a self-supervised contrastive learning approach to generate minutiae embeddings, improving fingerprint representation and matching. We first introduce a method to synthesize realistic latent fingerprints from rolled and plain images by applying ridge distortions, contrast shifts, blurring, noise, and document-based backgrounds. The resulting dataset includes reliable minutiae correspondences for effective training. Fingerprints are then represented as orientation-aligned, minutia-centered patches. A Siamese network trained with contrastive learning on these patches produces discriminative embeddings. Matching computes the mean cosine similarity between the embeddings of paired minutiae from candidate references selected by a matcher. Experiments on NIST SD27 and SD302, using a 20,473-print gallery, demonstrate rank-1 identification gains of 4.25 and 1.66 percentage points over prior work. It also consistently outperforms other synthetic latent generation baselines.

1. Introduction

Latent fingerprints often serve as critical evidence in criminal investigations and are routinely accepted in court proceedings [1]. Unlike rolled or slap fingerprints, which are deliberately collected under controlled conditions, latent prints are unintentionally deposited on object surfaces. As a result of their uncontrolled acquisition, latent fingerprints are frequently of much lower quality [2].

Latent fingerprint identification involves matching a low-quality, often partial query impression against an extensive gallery of reference prints. Noise, occlusions, misalignments, and unknown geometric distortions challenge this process. A typical pipeline includes enhancement, segmentation, feature extraction, and matching, all of which must operate reliably under these adverse conditions. Fig. 1 illustrates challenging latent examples from NIST SD27 alongside their corresponding reference prints.

Fingerprint representation methods can be broadly classified as global or local [1]. Global approaches, like DeepPrint [4], compute a single feature vector per image, whereas local methods describe multiple regions per fingerprint, typically centered at minutiae [5], to increase robustness against distortions and missing regions. Beyond patch-based descriptors, local methods may also encode geometric structures using Delaunay triangulation and its variants [6], capturing spatial configurations of minutiae.

A common approach to learning deep minutiae embeddings is to treat the problem as a classification task, where each minutia is assigned a unique identity and the network is trained using classification losses such as ArcFace [7–9]. Although effective in theory, this strategy requires large-scale datasets with multiple labeled samples per fingerprint to ensure meaningful intraclass and interclass variation, an impractical requirement given the difficulty of acquiring and annotating real latent fingerprints at scale.

An alternative strategy is to use contrastive learning frameworks, often implemented with Siamese networks, which learn to pull together embeddings of genuine minutiae pairs and push apart those of impostor pairs [10,11]. Although this approach reduces the need for large labeled datasets, most existing methods still rely on extensive, often private, datasets for training. Moreover, they typically optimize for Euclidean distance between minutiae embeddings, which may underutilize the semantic structure present in fingerprint patterns [12].

Both strategies also face a critical bottleneck: the need to establish reliable correspondences between minutiae in matching fingerprint pairs. Manual annotation of minutiae correspondences is infeasible at

* Corresponding author at: Institute of Computing - University of Campinas (Unicamp), Av. Albert Einstein, 1251, Campinas, 13083-852, Brazil.
E-mail addresses: a203758@dac.unicamp.br (A. Nobrega), i257199@dac.unicamp.br (I. Theodoro), pascual.figueroa@griaule.com (P. Figueroa), afalcao@ic.unicamp.br (A.X. Falcão).

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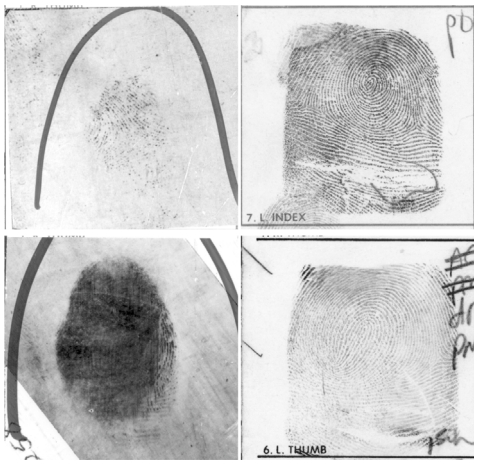


Fig. 1. Challenging examples of latent fingerprints and corresponding reference images from NIST SD27 [3] database.

scale, so most works rely on automated techniques, such as Minutiae Cylinder Codes (MCC) for representation and spectral clustering for matching, to generate pseudo-labels. These heuristics are inherently noisy and can propagate errors into the learning process, limiting the generalization and reproducibility of resulting models.

To address the limitations of existing embedding strategies, this paper proposes a self-supervised contrastive learning framework for latent fingerprint identification. Our approach relies on a local representation, where a fingerprint is modeled as a set of orientation-aligned patches centered on the detected minutiae. A Siamese neural network is trained to produce discriminative embeddings for these patches using a contrastive loss function. Specifically, the network is optimized to increase the cosine similarity between embeddings of genuine minutiae pairs, while reducing the similarity for impostor pairs. We deliberately adopt cosine similarity, rather than the commonly used Euclidean distance, because it better captures the angular and semantic relationships inherent in the patterns of the ridges of fingerprints [10].

A key component of our approach is the introduction of a synthetic latent fingerprint generator that eliminates the need for manual or algorithmic minutiae correspondence annotation. Starting from a private dataset of 6000 rolled and plain fingerprints, comparable in structure and to the reference images of the public NIST SD302 [13] dataset, we apply a series of carefully designed degradations, including geometric distortions, partial occlusions, illumination variation, blur, noise, and realistic document-based backgrounds, to simulate latent fingerprint conditions. Because each synthetic latent is derived from a known reference, minutiae correspondences are inherently available, requiring no additional labeling effort. Preprocessing steps such as enhancement, segmentation, and minutiae extraction are performed using FingerNet [14], allowing our contributions to focus entirely on robust representation learning and matching. To promote reproducibility and further research, we make our synthetic latent fingerprint generator publicly available.[1]

A preliminary version of this work was presented at SIBGRAPI 2024 [15]. In this extended version, we introduce several important improvements in both methodology and experimental validation. The primary enhancement lies in the development of a more realistic synthetic latent fingerprint generator, which now incorporates complex background textures from the RVL-CDIP dataset, comprising over 400,000 grayscale scans of historical litigation documents. These realistic degradations expose the network to a broader range of visual noise, enabling more effective learning of robust minutiae descriptors. Unlike the earlier version, this generator introduces degradations such as partial occlusions, ridge warping, lighting variation, and surface noise in a more structured and comprehensive way.

In addition to the improved generation pipeline, we conducted a significantly more extensive evaluation. The fingerprint gallery was expanded from 9067 to 20,473 reference impressions, substantially increasing the difficulty of the identification task. We also included the NIST SD302 latent subset in our evaluation to validate generalization beyond the SD27 benchmark. Moreover, we performed a direct comparison with recent synthetic latent generation methods proposed by Guan [16] and Grosz and Jain [9], integrating their generators into our pipeline and retraining the minutiae descriptor network accordingly. Our method outperformed these alternatives, achieving 4.25 and 1.66 percentage points improvement in rank-1 identification over our previous work for SD27 and SD302, respectively.

Finally, we conducted a comprehensive ablation study to quantify the contribution of each proposed transformation (e.g., ridge distortion, blur, occlusion, background composition) to the quality of the learned descriptors. These results demonstrate the effectiveness of each degradation component in simulating real latent fingerprint variability.

The main contributions of this extended paper are:

- The introduction of a significantly more realistic synthetic latent fingerprint generator.
- Extensive experimental improvements, including an ablation study, larger-scale evaluation on a 20,473-reference gallery, experiments on NIST SD302 latents, and comparisons with state-of-the-art synthetic latent generation techniques.
- A measured improvement of 4.25 percentage points increase in rank-1 identification rate on NIST SD27 compared to our prior work, and 1.66 percentage points increase on NIST SD302, with a 10.10 percentage points margin over a commercial baseline (Griaule SDK).

The rest of the paper is organized as follows: Section 2 reviews related work on minutiae descriptor learning and synthetic latent fingerprint generation. Section 3 describes our improved synthetic data generator and self-supervised representation learning approach. Section 4 presents detailed experimental results. Finally, Section 5 concludes the paper and outlines future research directions.

2. Related work

This study investigates self-supervised deep minutiae descriptors under limited data, emphasizing: (1) realistic synthesis of forensic-style latent fingerprints, and (2) contrastive learning to capture semantic minutiae similarity. A review of prior work provides context and highlights key limitations.

2.1. Realistic synthetic latent fingerprint generation

Due to the scarcity of large-scale annotated latent fingerprint datasets, synthesizing realistic latent fingerprints has become essential for training deep learning models.

Zhu et al. introduced FingerGAN [17], which simulates latent fingerprints by degrading rolled impressions with plastic distortions and speckle noise, using Total Variation decomposition to extract background noise from NIST SD27. This reuse of SD27, however, may introduce data leakage and compromise the fairness of evaluation.

A recent advance in a similar direction is FPGAN-Control [18], which introduces an identity-preserving generative framework capable of producing multiple impressions of the same synthetic fingerprint identity under controllable appearance factors. Remarkably, this is the first approach to demonstrate that recognition models can be trained using only synthetic fingerprints, achieving competitive accuracy comparable to that of real-data training. Nevertheless, FPGAN-Control is

[1] https://github.com/LIDS-UNICAMP/synthetic_fingerprint_generation.

restricted to the reference fingerprint domain, which is comparatively less challenging and has been widely explored. Our work follows a similar philosophy of leveraging synthetic variability to improve recognition, but in the more demanding latent fingerprint scenario.

In addition, a recent contribution also explored training purely on synthetic data, but in this case to learn a multi-task GAN for latent fingerprint enhancement [19]. Similar to our approach, it simulated diverse latent conditions, though with the specific goal of enhancement rather than representation learning.

Cunha et al. [20] created synthetic latents from NIST SD300 and SFinGe-generated prints by applying Gaussian blur, noise, salt-and-pepper artifacts, and contrast reductions. While effective for enhancement training, their method omits occlusion simulation and complex backgrounds.

Kriangkhajorn et al. [21] enhanced diversity by blending clean patches from NIST SD14 with 360 texture images from the DTD dataset. This introduced realistic background variation, but did not include ridge-specific distortions such as pressure effects or occlusions.

Grosz and Jain [9] addressed latent fingerprint generation through multiple data augmentation techniques, including Gaussian noise, Gaussian blurring, downsampling, partial occlusions, and contrast adjustments applied to rolled and plain fingerprint images. These synthetic latents were used to train enhancement networks. Nevertheless, their approach lacks the complex and diverse background textures found in real latent fingerprints, restricting the realism of their generated dataset.

Guan et al. [16] proposed a synthetic fingerprint generation strategy tailored explicitly to partial fingerprint scenarios. Their method involves a rich set of handcrafted transformations designed to simulate realistic degradations. These include mirror flipping, rotation, and grayscale inversion to enhance diversity; Gaussian blur and noise with varying kernels and variances to model sensor noise; morphological operations such as dilation and erosion to emulate wet and dry fingerprint conditions; and the addition or subtraction of random white blobs to simulate fingerprint-specific contamination, regional pollution, or weak impressions. These transformations are randomly combined to generate synthetic latents.

While Guan's approach is effective in introducing geometric and intensity-based distortions, it still lacks high-fidelity simulation of surface artifacts and realistic background variation, which are crucial for training models that are robust to the complex noise patterns present in latent fingerprints captured in real-world crime scenes.

In our experiments (Section 4.3), we incorporate both Guan's and Grosz's synthetic generators into our evaluation pipeline. We use their respective generators to create training data, keeping all other aspects of our pipeline, network architecture, contrastive learning setup, and gallery size identical.

2.2. Learning minutiae descriptors

Deep-learning-based minutiae descriptors have recently gained prominence for latent fingerprint matching, primarily due to their robustness against noise and distortion. Still, most state-of-the-art methods require extensive annotated data or rely on private datasets.

Grosz and Jain [9] proposed combining local and global fingerprint embeddings, trained with a minutiae classification metric, to enhance the accuracy of latent-to-rolled fingerprint matching. Their approach effectively leveraged local and global embeddings in a combined matching strategy, achieving a state-of-the-art identification rate on NIST SD27. However, their training process remains dependent on large-scale private datasets, limiting reproducibility and broader applicability.

Öztürk et al. introduced MinNet [8], a minutiae patch embedding network trained with weakly supervised correspondences from matched rolled and slap fingerprints. It jointly optimizes spatial and angular distributions of neighboring minutiae and ridge flow. Despite its high accuracy, MinNet still depends on reliable, albeit weakly labeled, correspondences, limiting scalability and ease of dataset preparation.

Pan et al. [22] proposed the Dense Minutiae Descriptor (DMD), which generates 3D minutiae embeddings from patches aligned to their central minutiae, along with segmentation maps to ensure validity in foreground regions. While DMD improves interpretability and performance, it relies on strongly labeled correspondences obtained through careful matching, requiring substantial preprocessing and high-quality labels—often difficult to obtain for latent fingerprints.

Our proposed self-supervised strategy departs from existing approaches by entirely removing the need for explicit minutiae correspondence labels. Instead, we use contrastive learning to implicitly encode semantic similarity directly from pairs of realistically synthesized latent fingerprints, achieving robust embeddings trained exclusively on publicly available datasets. This approach significantly simplifies training, reduces labeling errors, and enhances generalization.

3. Method

3.1. Deep minutiae descriptor network

We train our minutiae embedding network using a contrastive self-supervised learning framework inspired by SimCLR [23], which aims to bring representations of similar samples closer while pushing dissimilar ones apart. Unlike the original SimCLR formulation, where both views are augmented versions of the same sample, our approach considers one original patch and one augmented view, reflecting the nature of minutiae correspondence in fingerprint matching. Specifically, given a high-quality fingerprint patch p_i and its corresponding synthetic latent fingerprint patch $\tilde{p}_i$, we create embedding vectors using a Siamese neural network architecture. The network processes two input patches simultaneously, optimizing a loss function that encourages representations from genuine pairs (i.e., patches originating from the same minutia) to remain close, while ensuring embeddings from impostor pairs (i.e., patches from different minutiae) are distant. Fig. 2 illustrates this process.

Unlike previous approaches that rely on extensive labeled or private datasets [8,9], our framework uses only synthetically generated latent fingerprints and their corresponding high-quality reference fingerprint images, effectively addressing data availability challenges.

Our embedding extraction network follows the Siamese architecture employed in our previous work [15]. We adopt a ResNet18 backbone due to its balanced performance and computational efficiency. Both branches of the Siamese network share weights, ensuring consistency in embedding extraction. To train our network effectively without explicit labels, we utilize a self-supervised contrastive loss, shown in Eq. (1).

$$\mathcal{L}_{NTXent}(z_i, \tilde{z}_i) = -\log \frac{exp(sim(z_i, \tilde{z}_i)/\tau)}{\sum_{k=1}^{2N} 1_{[k \neq i]} exp(sim(z_i, z_k)/\tau)}, \tag{1}$$

where z_i and $\tilde{z}_i$ are the embeddings extracted by the network, $sim(a, b)$ is the cosine similarity function, τ is a temperature parameter, and $1_{[k \neq i]}$ is an indicator function that is 1 if $k \neq i$ and 0 otherwise. Here, N denotes the number of distinct minutiae pairs within a batch, resulting in $2N$ augmented samples used for contrastive learning.

3.2. Synthetic latent generation

We designed a synthetic latent fingerprint generator that reproduces key visual degradations commonly observed in forensic scenarios, such as partial occlusions, ridge deformations, lighting inconsistencies, and background interference.

Our generator sequentially applies a diverse set of image processing transformations, each targeting a specific type of distortion. These include morphological ridge modifications, contrast manipulation, blur, noise, and background blending, among others. Fig. 3 illustrates examples of synthetic generated fingerprints. The clean reference print corresponding to each synthetic version is also shown for comparison.

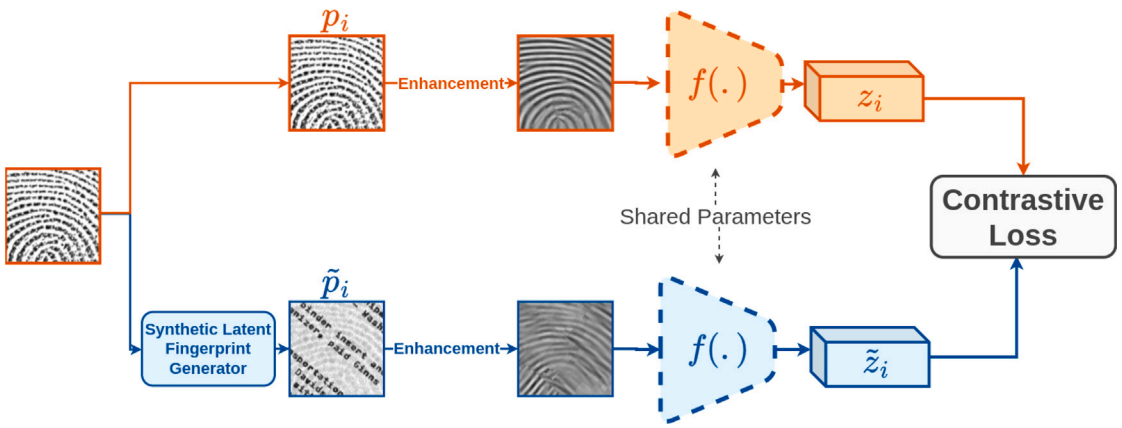


Fig. 2. Illustration of the proposed method pipeline. A reference fingerprint patch is transformed into a synthetic latent fingerprint patch through our enhanced synthetic latent generator. Both patches are passed through a Siamese network architecture to extract deep minutiae embeddings, which are subsequently compared using a contrastive loss function.

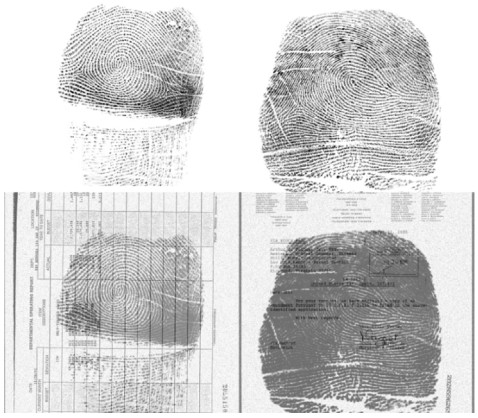


Fig. 3. Examples of generated synthetic latents. Each column shows a clean reference fingerprint (top) and its transformed version (bottom).

The synthetic latent generator applies a series of degradations to emulate real-world forensic conditions. To simulate pressure variation during fingerprint deposition, morphological operations (dilation or erosion) are applied with 50% probability using a square kernel ($k \in \{2, 3, 4\}$) to modify ridge thickness. Lighting inconsistencies are modeled via gamma correction, with values sampled from $[0.3, 0.7]$ or $[3.0, 4.0]$, mimicking under, or overexposure. Motion blur and sensor noise are simulated by applying a Gaussian blur (kernel sizes 3 or 5) and multiplicative speckle noise with variance in $[0.1, 0.3]$.

To emulate low-resolution acquisition, average pooling is performed with block sizes from $\{2, 3\}$, and this is often combined with gamma adjustments to represent optical degradation. Partial impressions and occlusions are introduced via parabolic and linear masks. Parabolic masks have a thickness between 7 and 15 pixels and opacity between 0.3 and 0.9, while 3–7 linear occlusions are added with similar parameters. Finally, to simulate surface texture variation, each fingerprint is blended with a background sampled from the RVL-CDIP dataset using an opacity factor $\alpha \in [0.2, 0.8]$, introducing realistic document-like noise into the image.

The ranges of the augmentation parameters were determined empirically through an iterative process. Candidate ranges were progressively adjusted until the generated samples visually resembled realistic latent impressions in terms of ridge distortions, background interference, and overall degradation levels. This qualitative optimization was complemented by our ablation study in Section 4.4, which quantifies the impact of each transformation type on identification performance.

Table 1
Architecture of the Siamese minutiae descriptor network.

Component	Configuration
Input	$1 \times 192 \times 192$ grayscale patch
Backbone	ResNet-18
Output of backbone	512-dimensional feature vector
Projection head	ReLU + Linear($512 \rightarrow 32$)
Final descriptor	32-dimensional embedding

Following the application of these transformations, we crop 192×192 patches centered at each minutia point and align them to the horizontal axis based on the minutia orientation. This patch size was selected based on findings from the end-to-end latent fingerprint search study by Cao et al. [7], which demonstrated that larger patch sizes improve descriptor robustness and preserve sufficient ridge structure for reliable matching. Additionally, to simulate errors from automatic minutiae extractors, we apply random translations (up to 30 pixels) and small angular rotations (up to 15°) to each patch.

Finally, as presented in our ablation study 4.4, we evaluate the individual contribution of each transformation to the effectiveness of the learned minutiae descriptors, demonstrating their complementary role in improving generalization.

3.3. Network architecture and implementation details

We adopt a Siamese network to learn minutiae-centered embeddings. Each branch of the network processes a 192 × 192 grayscale patch using a ResNet-18 backbone followed by a non-linear projection head. The architecture is summarized in Table 1.

We train the model using the Adam optimizer with a learning rate of 0.0001 and weight decay of 0.1. A MultiStep learning rate scheduler is applied with milestones at epochs 30 and 150, and a decay factor $\gamma = 0.1$. Training is conducted using a batch size of 2048 on 4 NVIDIA H100 GPUs. The model is trained for up to 200 epochs with early stopping based on validation loss. A temperature parameter $\tau = 0.1$ is used in the contrastive loss function.

3.4. Minutiae matching using deep embeddings

Once minutiae embeddings have been extracted, the fingerprint matching procedure begins by establishing minutiae correspondences between the latent and reference fingerprint images. Given two sets of minutiae embeddings $A = \{a_1, a_2, \ldots, a_M\}$ from a latent fingerprint and $B = \{b_1, b_2, \ldots, b_N\}$ from a reference fingerprint, we first apply Griaule's

minutiae matcher to identify candidate minutiae pairs based on spatial and angular proximity.

For each identified minutiae pair (a_i, b_j), we calculate their similarity score using the cosine similarity between their respective deep embedding vectors using Eq. (2).

$$sim(a_i, b_j) = \frac{a_i^T b_j}{\|a_i\| \|b_j\|}. \tag{2}$$

The global similarity score $S(A, B)$ between the latent and reference fingerprints is then computed by averaging the similarity scores of all corresponding minutiae pairs (Eq. (3)).

$$S(A, B) = \frac{1}{P} \sum_{a_i, b_j \in P} sim(a_i, b_j), \tag{3}$$

where P represents the set of minutiae correspondences established in the first step.

4. Experiments

4.1. Datasets

To thoroughly evaluate our proposed approach, we used a combination of public and private fingerprint datasets for training, testing, and large-scale gallery evaluation.

For training, we employed a private dataset containing 6000 rolled and plain fingerprint impressions from distinct individuals. This dataset closely mirrors the characteristics of the publicly available NIST SD302 dataset [13] in terms of image quality and acquisition conditions, but it remains separate to preserve SD302 for evaluation purposes. From this private dataset, 80% of the generated minutiae patches were used for training the embedding network, while the remaining 20% served as a validation set during training.

For testing, we used two standard benchmark datasets. NIST SD27 [3], which includes 258 latent fingerprints and their corresponding rolled reference impressions. This dataset is widely used for evaluating latent fingerprint identification methods under challenging conditions. NIST SD302 [13], which we used as a second evaluation benchmark.

To simulate a realistic large-scale identification scenario, we used a private fingerprint gallery comprising 20,473 reference impressions.

4.2. Deep minutiae embedding training

We evaluated our deep minutiae embedding network by analyzing cosine similarity distributions between genuine and impostor minutiae pairs. Ideally, genuine pairs yield scores close to 1, while impostors approach 0.

Genuine correspondences were established using the Griaule SDK on the NIST SD27 test set and a held-out 20% subset of reference images from the NIST SD302 dataset. This setup does not involve latent-to-latent matching, making it a comparatively more straightforward subset than SD27. To reduce label noise, we retained only matches with SDK similarity scores above 0.9.

Fig. 4 presents histograms illustrating the distributions of embedding similarity scores for genuine and impostor minutiae pairs in both the NIST SD27 latent test set and the NIST SD302 reference test set. For both sets, we compute accuracy by applying a threshold on the similarity score to classify pairs. The thresholds shown in the figure are selected to maximize classification accuracy between genuine and impostor pairs. On the NIST SD302 reference test set, we observed near-perfect discrimination, achieving accuracy above 99%. This outcome highlights the network's strong ability to differentiate between true correspondences and mismatches under controlled conditions.

On the more challenging NIST SD27 test set, the network achieved approximately 90% accuracy, a 6.40% gain over our previous work [15].

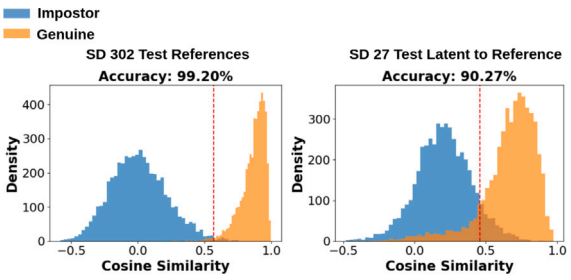


Fig. 4. Distribution of similarity scores between genuine and impostor minutiae pairs for test datasets. The clear separation highlights the discriminative power of our deep minutiae embeddings.

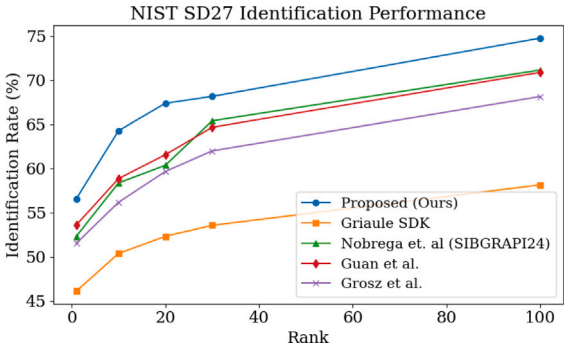


Fig. 5. Identification results on NIST SD27. Our method consistently outperforms the commercial Griaule SDK baseline and alternative synthetic generation methods across all ranks.

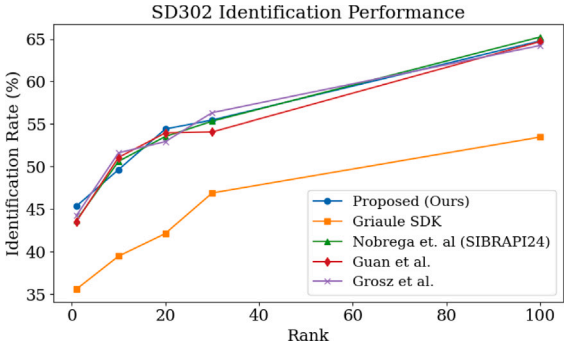


Fig. 6. Identification results on NIST SD302. As a latent-to-reference benchmark.

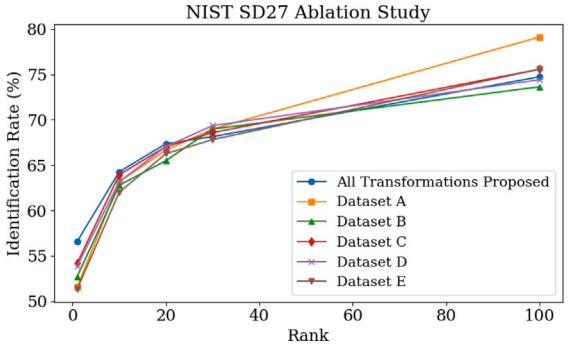


Fig. 7. Ablation study results on NIST SD27. Each curve corresponds to a version of the synthetic training set with one transformation component removed (Datasets A–E). The full method ("Proposed") includes all transformations.

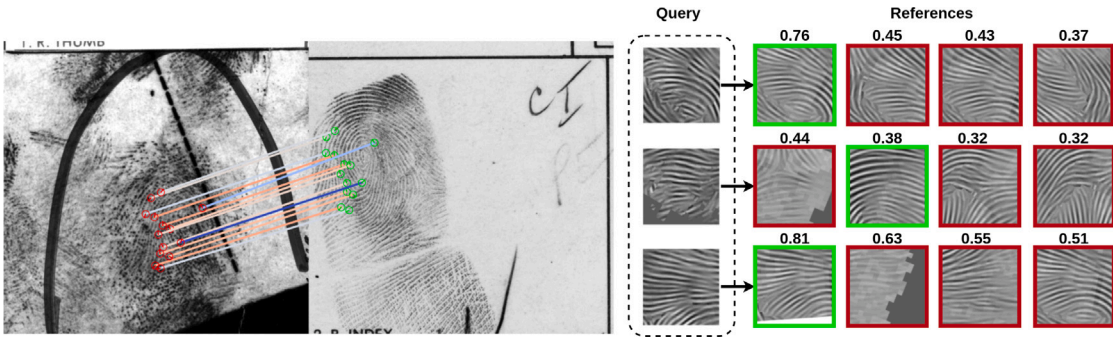


Fig. 8. Examples of qualitative minutiae matches. Left: latent fingerprint with detected minutiae; Center: reference print; Right: minutiae patches from the query and their corresponding top-1 matches, with cosine similarity scores. Green borders indicate correct matches, while red borders indicate incorrect matches.

4.3. Identification results

To evaluate the practical effectiveness of our method in realistic forensic scenarios, we conducted large-scale latent fingerprint identification experiments on two standard benchmarks: NIST SD27 and NIST SD302. For each latent fingerprint in these datasets, we computed similarity scores against all entries in a gallery of 20,473 reference impressions, using our cosine-based minutiae matching strategy described in Section 3.3. Identification performance was measured using rank-N accuracy, which reflects the percentage of queries whose correct match appears within the top-N ranked results.

We compared our proposed approach with several baselines. The Griaule SDK serves as our commercial baseline. We also evaluated the previous SIBGRAPI version of our method, which used a less sophisticated synthetic generation pipeline. In addition, we incorporated synthetic latent generators proposed by Guan and Grosz and Jain into our pipeline, retraining our descriptor using the same architecture and training configuration, and isolating the effect of the generator.

Figs. 5 and 6 report the identification results on NIST SD27 and NIST SD302, respectively. On NIST SD27, our method achieves a rank-1 accuracy of 56.57%, outperforming the Griaule SDK baseline by over 10 percentage points and all other synthetic-generation-based methods across all evaluated ranks. On NIST SD302, a latent-to-reference identification benchmark, our method also demonstrates superior generalization, yielding higher accuracy than both the SIBGRAPI version and the methods trained using synthetic latents from Guan and Grosz.

While our discussion has emphasized rank-1 results, the full rank-N curves in Figs. 5, 6, and 7 provide additional insights. On NIST SD27, our method consistently outperforms baselines across ranks. On NIST SD302, although it yields the best rank-1 accuracy, other methods slightly surpass it at higher ranks. A similar pattern appears in the ablation study, where using all transformations maximizes rank-1 performance, but some reduced variants achieve comparable results at larger ranks. These findings indicate that our generator is particularly effective at improving top-rank precision, but further refinements are needed to extend these improvements consistently across higher-rank retrieval scenarios.

4.4. Ablation study

To better understand the individual contribution of each transformation in our synthetic latent fingerprint generator, we conducted an ablation study using the NIST SD27 dataset. We created five synthetic training datasets (labeled A through E), each omitting one transformation component while keeping all others unchanged. We then retrained our descriptor model on each modified dataset and repeated the identification experiments against the 20K gallery.

We evaluated five ablation variants of our synthetic dataset. In **Dataset A**, we removed occlusion modeling by excluding both parabolic and linear masks, which simulate partial impressions and surface obstructions. **Dataset B** omits ridge thickness variation by disabling dilation and erosion operations that mimic pressure-based distortions. **Dataset C** excludes illumination variation by removing gamma contrast adjustments and downsampling, both of which help simulate lighting inconsistencies and poor acquisition quality. **Dataset D** removes image degradation effects, specifically Gaussian blur and speckle noise, which are used to model defocus and sensor noise. Finally, **Dataset E** disables background texture blending, excluding the realistic overlays from the RVL-CDIP dataset that simulate complex surface textures. Fig. 7 shows the identification performance for each variant.

The results show that each transformation plays a meaningful role in descriptor quality. The most significant drop in rank-1 identification occurs in Dataset E (–5.26 percentage points), where background textures were removed, and in Dataset A (–5.04 percentage points), where occlusions were excluded. This suggests that simulating surface clutter and partial impressions is particularly important for robust matching.

4.5. Qualitative results and discussions

Fig. 8 illustrates examples of minutiae matches between a latent fingerprint (left) and its corresponding reference print (center). On the right, we show query minutiae patches alongside the top-scoring matches from the gallery, with similarity scores. Matches with high similarity correspond to visually aligned ridge patterns, while those involving low-quality query patches result in decreased similarity.

While the network successfully identifies several accurate minutiae correspondences, failure cases still occur, particularly when the query patch is heavily degraded. As shown in Fig. 8, the middle query patch retrieves an incorrect top-1 match.

Our generator simulates latent fingerprints over paper-like backgrounds from the RVL-CDIP dataset. In practice, however, latents appear on diverse surfaces — such as glass, metal, plastic, fabric — each introducing unique reflectance and texture artifacts. Real-world degradations are challenging to replicate with standard filters, highlighting the need for broader and more realistic degradation modeling.

It is also important to note that this work relies on FingerNet for enhancement, segmentation, and minutiae extraction. If an alternative pipeline were used for these preprocessing steps, the quality and distribution of the resulting minutiae patches could change significantly. In such cases, the proposed descriptor network may need to be retrained to maintain performance.

5. Conclusion

This work tackled latent fingerprint identification under limited data availability by introducing a more realistic synthetic latent fingerprint generator. Our generator combines morphological ridge distortions, diverse degradations, and real background textures from the

RVL-CDIP dataset, yielding synthetic latents that better reflect forensic conditions—though predominantly limited to paper-based surfaces.

Using this data, we trained deep minutiae embeddings via a self-supervised contrastive learning framework, removing the need for manual annotation while enabling robust and discriminative representations.

Experiments on NIST SD27 confirmed the effectiveness of our approach, with a rank-1 identification improvement of 4.25 percentage points over our prior work and performance comparable to a leading commercial matcher (Griaule SDK). Additionally, on the NIST SD302 benchmark, our method achieved a rank-1 improvement of 1.66 percentage points, further validating its generalization ability in latent-to-reference identification scenarios.

To foster future research, we publicly release our improved synthetic latent generator, creating labels to train models for tasks such as enhancement and minutiae extraction.

CRediT authorship contribution statement

Andre Nobrega: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Ilan Theodoro:** Writing – review & editing, Validation, Investigation, Funding acquisition, Conceptualization. **Pascual Figueroa:** Writing – review & editing, Validation, Supervision, Conceptualization. **Alexandre Xavier Falcão:** Writing – original draft, Validation, Supervision, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] D. Valdes-Ramirez, et al., A review of fingerprint feature representations and their applications for latent fingerprint identification: Trends and evaluation, IEEE Access 7 (2019) 48484–48499, http://dx.doi.org/10.1109/ACCESS.2019.2909497.

[2] D. Maltoni, D. Maio, A.K. Jain, J. Feng, Handbook of Fingerprint Recognition, Springer, 2022.

[3] M.D. Garris, R.M. McCabe, NIST Special Database 27:, Technical Report, National Institute of Standards and Technology, NIST, 2000, http://dx.doi.org/10.6028/nist.ir.6534.

[4] J.J. Engelsma, K. Cao, A.K. Jain, Learning a Fixed-Length fingerprint representation, IEEE Trans. Pattern Anal. Mach. Intell. 43 (6) (2021) 1981–1997, http://dx.doi.org/10.1109/TPAMI.2019.2961349.

[5] A. Vij, A. Namboodiri, Learning minutiae neighborhoods: A new binary representation for matching fingerprints, in: 2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2014, pp. 64–69, http://dx.doi.org/10.1109/CVPRW.2014.15.

[6] G. Bebis, T. Deaconu, M. Georgiopoulos, Fingerprint identification using Delaunay triangulation, in: Proceedings 1999 International Conference on Information Intelligence and Systems (Cat. No.PR00446), 1999, pp. 452–459, http://dx.doi.org/10.1109/ICIIS.1999.810315.

[7] K. Cao, D.-L. Nguyen, C. Tymoszek, A.K. Jain, End-to-End latent fingerprint search, IEEE Trans. Inf. Forensics Secur. 15 (2020) 880–894, http://dx.doi.org/10.1109/TIFS.2019.2930487.

[8] H.I. Ozturk, B. Selbes, Y. Artan, MinNet: Minutia patch embedding network for automated latent fingerprint recognition, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, CVPRW, 2022, pp. 1626–1634, http://dx.doi.org/10.1109/CVPRW56347.2022.00169.

[9] S. Grosz, A. Jain, Latent fingerprint recognition: Fusion of local and global embeddings, IEEE Trans. Inf. Forensics Secur. PP (2023) 1, http://dx.doi.org/10.1109/TIFS.2023.3314207.

[10] F. Zhang, S. Xin, J. Feng, Combining global and minutia deep features for partial high-resolution fingerprint matching, Pattern Recognit. Lett. (ISSN: 0167-8655) 119 (2019) 139–147, http://dx.doi.org/10.1016/j.patrec.2017.09.014, Deep Learning for Pattern Recognition.

[11] S. Gu, J. Feng, J. Lu, J. Zhou, Latent fingerprint registration via matching densely sampled points, 2020, CoRR abs/2005.05878.

[12] R. Balestriero, M. Ibrahim, et al., A cookbook of Self-Supervised learning, 2023, URL https://arxiv.org/abs/2304.12210.

[13] G. Fiumara, P. Flanagan, J. Grantham, K. Ko, K. Marshall, M. Schwarz, E. Tabassi, B. Woodgate, C. Boehnen, NIST Special Database 302:, Technical Report, National Institute of Standards and Technology, 2019, http://dx.doi.org/10.6028/nist.tn.2007.

[14] Y. Tang, F. Gao, J. Feng, Y. Liu, FingerNet: An unified deep network for fingerprint minutiae extraction, in: 2017 IEEE International Joint Conference on Biometrics, IJCB, IEEE, 2017, pp. 108–116.

[15] A. Nóbrega, et al., Improving local latent fingerprint representations under data constraints, in: 2024 37th SIBGRAPI Conference on Graphics, Patterns and Images, 2024, pp. 1–6, http://dx.doi.org/10.1109/SIBGRAPI62404.2024.10716328.

[16] X. Guan, et al., Joint identity verification and pose alignment for partial fingerprints, IEEE Trans. Inf. Forensics Secur. 20 (2025) 249–263, http://dx.doi.org/10.1109/TIFS.2024.3516566.

[17] Y. Zhu, X. Yin, J. Hu, FingerGAN: A constrained fingerprint generation scheme for latent fingerprint enhancement, IEEE Trans. Pattern Anal. Mach. Intell. 45 (7) (2023) 8358–8371, http://dx.doi.org/10.1109/TPAMI.2023.3236876.

[18] A. Shoshan, N. Bhonker, E.B. Baruch, O. Nizan, I. Kviatkovsky, J. Engelsma, M. Aggarwal, G. Medioni, FPGAN-Control: A Controllable Fingerprint Generator for Training with Synthetic Data , in: 2024 IEEE/CVF Winter Conference on Applications of Computer Vision, WACV, IEEE Computer Society, Los Alamitos, CA, USA, 2024, pp. 6055–6064, http://dx.doi.org/10.1109/WACV57701.2024.00596, URL https://doi.ieeecomputersociety.org/10.1109/WACV57701.2024.00596.

[19] R. Pramukha, P. Akhila, S.G. Koolagudi, End-to-end latent fingerprint enhancement using multi-scale Generative Adversarial Network, Pattern Recognit. Lett. (ISSN: 0167-8655) 184 (2024) 169–175, http://dx.doi.org/10.1016/j.patrec.2024.06.022, URL https://www.sciencedirect.com/science/article/pii/S0167865524001910.

[20] N.D.S. Cunha, H.M. Gomes, L.V. Batista, Residual M-net with Frequency-Domain loss function for latent fingerprint enhancement, in: 2022 35th SIBGRAPI Conference on Graphics, Patterns and Images, Vol. 1, 2022, pp. 198–203, http://dx.doi.org/10.1109/SIBGRAPI55357.2022.9991793.

[21] S. Kriangkhajorn, K. Horapong, V. Areekul, Spectral filter predictor for progressive latent fingerprint restoration, IEEE Access 12 (2024) 66773–66800, http://dx.doi.org/10.1109/ACCESS.2024.3397729.

[22] Z. Pan, Y. Duan, X. Guan, J. Feng, J. Zhou, Latent fingerprint matching via dense minutia descriptor, in: 2024 IEEE International Joint Conference on Biometrics, IJCB, 2024, pp. 1–10, http://dx.doi.org/10.1109/IJCB62174.2024.10744445.

[23] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: H.D. III, A. Singh (Eds.), Proceedings of the 37th International Conference on Machine Learning, in: Proceedings of Machine Learning Research, Vol. 119, PMLR, 2020, pp. 1597–1607.